Computer assignment 3
ALM-analysis

Begränsad delning

Introduction

In this assignment you will implement an ALM-model of a life insurance company to calculate risk capital, study the impact of various investment strategies and perform a stress test. The model will make use of the asset scenarios produced in assignment 2.

Policy data:

Use the same policy data as in assignment 2. Set the prognosis rate and the transfer rate to 2%.

Asset portfolio:

The investment universe consists of the ten stocks and the five rollover bonds, corresponding to the zero-coupon rates, from assignment 2. The initial portfolio value and 2nd order reserve is equal to V(0) = 2100000.

Scenarios:

You are provided with two scenario files:

- **OneScenario.csv:** containing one scenario over 720 monthly time periods where each stock grows by 5% each year and a flat discount rate of 3%, and
- **ScenarioFile.csv:** with 2000 scenarios over 120 monthly time periods.

The scenario files are structured as (where Z5-Z30 denote zcb prices and Y5-Y30 denote zero rates):

Scenario	Period	BA	CAT	KO	DIS	JPM	...	Z20	Z30	Y5	Y10	Y15	Y20	Y30
1	0	218.500000	573.882996	68.734924	113.917114	318.891754	...	100.000000	100.000000	0.036592	0.042127	0.046299	0.048846	0.050492
1	1	201.806264	511.631351	67.604561	84.400383	261.155980	...	112.753887	119.921198	0.028545	0.035861	0.040478	0.043023	0.044560
1	2	167.402263	430.475457	64.566459	73.576724	215.484455	...	122.635956	129.312505	0.021607	0.029339	0.035220	0.038985	0.042164
1	3	153.857659	470.627803	70.213817	75.248426	202.368262	...	122.701107	126.096190	0.021065	0.029346	0.035315	0.039122	0.043124
1	4	160.816784	506.729865	68.129273	73.558024	231.343360	...	128.025176	138.885664	0.019928	0.028208	0.033842	0.037153	0.040014

1. The state model

Implement the state model for the policy and plot the populations (PP, PO, TR, DE) for each period 50 years (600 months) into the future. Use the area plot to visualize the size of the populations over time.

2. Implement an ALM-model

Implement an ALM-model that in each scenario, and in each time period, calculates the:

- expected 1st-order reserve (G),
- expected 2nd-order reserve (V),
- value of the assets,
- technical provisions (only BE) and basic own funds (BOF)

Test your ALM-model by running it for a single scenario (OneScenario.csv) where the investment portfolio is equally allocated among the stocks, but no bond investment.

a) Plot the evolution of the expected reserves, assets, and BE in the same plot for 30 years.
b) Include a table with the values of the above items at every fifth year.
c) Calculate $SCR_{longevity}$ according to the Solvency 2 standard model.

3. Estimation of risk capital

Use simulation to estimate the SCR, which we here take as the 1-year VaR for the loss in basic own funds at the 95% confidence level[1], and plot also the empirical distribution of BOF_1 as a histogram with fixed x-limits: (-500 000, 3 000 000) and 50 bins, for the investment strategies below:

a) BH(0/100), where you are entirely invested in the Z15 bond.
b) BH(50/50), where the initial stock investment is equally distributed among the ten stocks and the bond investment is entirely invested in the Z5 bond.
c) CM (50/50), where the stock investment is equally distributed among the ten stocks and the bond investment is entirely invested in the Z5 bond.

[1] *Recall that 99.5% is used in Solvency II, but we choose 95% to get a stable estimate since we are only using 2000 scenarios.*

d) CM (50/50), where the stock investment is equally distributed among the ten stocks and the bond portfolio consists of the Z5 and Z30 bonds[2] and having duration equal to the future 1st order cashflow in each period.
e) CPPI(2), where the stocks[3] have equal weight and the bond portfolio consists of the Z5 and Z30 bonds and having duration equal to the future 1st order cashflow in each period.

Discussion: Comparing the strategies, explain to what extent the SCR results support the idea that asset-liability management can reduce solvency risk?

4. ALM-analysis

Perform a balance sheet simulation over a 10-year projection period to study the impact of the investment strategies in 3 (that you have chosen to implement) on basic own funds and the probability of insolvency.

- For each strategy, first calculate the median of own funds after ten years (in the median calculation you should only include the scenarios which never become insolvent during the simulation) and then scatter plot all the strategies in the same ”median – insolvency probability” diagram (with the median of basic own funds on the y-axis and the insolvency probability on the x-axis). Include also a table with the medians and probabilities.
- Also, for scenarios 76 and 82 in the scenario file, plot the evolution of the assets, expected 1st- and 2nd order reserve, BE and basic own funds, in the same plot, for each of the investment strategies (up to ten plots total).
- Plot the total stock and the total bond allocation for CPPI in the two scenarios. Use an area plot.

Discussion: Which strategy would you choose if you were the CFO of the company? Comment on the differences in performance of the different strategies in the two selected scenarios?

5. Stress test

Perform a stress test based on historical data for investment strategies a) and b). Specifically, re-run year 2008 and determine its effect on the BOF, that is calculate the change in BOF (ΔBOF) during the year for both

[2] *If the duration is less than 5 years, let the bond portfolio be fully invested in the 5Y bond, and if the duration is greater than 30 years, let the bond portfolio be fully invested in the 30Y bond.*
[3] *Make sure that the total stock investment is less than the total asset value in each period, you are not allowed to short bonds.*

strategies. To do this you need to construct a scenario file, based on market_data.csv, to be fed into your ALM-model:

1. calculate the monthly returns during 2008 (also for the interest rates)
2. apply these returns to the stock prices and interest rates at the end of 2025
3. determine the rollover zcb prices

Include your constructed scenario file in the report.

Presentation of results

Strategy	SCR	Median BOF	Insolvency probability	ΔBOF (only for 8)
BH(100/0)				
BH(50/50)				
CM(50/50)				
CM(50/50)-duration				
CPPI(2)				

Grading

These are the requirements for the grades 3, 4 and 5:

Problem	3	4	5
1	All	All	All
2	All	All	All
3	a,b	a-d	All
4	a,b	a-d	All
5	-	All	All